A Lean and Robust Scale-Up Strategy for an Esterification Process in the Oleochemical Industry

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Background:

Esterification is a key reaction in oleochemical manufacturing for producing specialty esters. Scaling these reactions from laboratory to pilot scale often introduces safety, quality, and operability challenges that are not evident at small scale.

Problem Statement:

Lack of a structured scale-up framework leads to uncontrolled risks, variable product quality, and operational inefficiencies during lab-to-pilot transfer of esterification processes.

REACTION: GLYCEROL+ STEARIC ACID → GLYCEROL MONOSTEARATE + WATER

TYPE OF REACTION: Batch-direct esterfication

CATALYST: P-TOLUENE SULFONIC ACID

SCALE AND SCOPE LADDER

- **Lab scale :** 500 mg to 1kg
- **Kilogram scale:** 5-10kg
- **Pilot scale:** 100kg

Critical Process Parameters & Quality Control

CRITICAL PROCESS PARAMETERS (CPPs) – OPERATING WINDOW & IMPACT

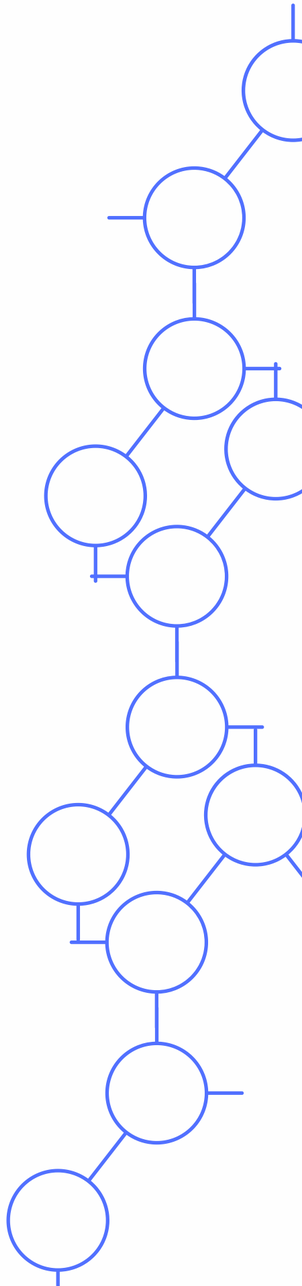
CPP Aspect	Recommended Operating Range / Control	Process Impact
Temperature	130–150 °C	↑ Temperature → ↑ conversion & purity, ↓ acid value; ↑ exessive T → colour fomation risk
Acid : Alcohol Ratio	1 : 1.2 – 1.5 (molar)	Excess glycerol shifts equilibrium toward ester formation
Catalyst Loading	1–2 wt% (on stearic acid)	↑ Catalyst → faster reaction; excess → darker colour
Water Removal Rate	Match lab profile ±10%	Poor removal → equilibrium limitation & conversion plateau
Agitation / Mixing	Maintain constant power per unit volume	Inadequate mixing → hot spots, side reactions, ↑ colour
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Quality Control

QC Aspect	Lab (1 kg)	Kilo (10 kg)	Pilot (100 kg)
Scale	End-point only (4 h)	4 timepoints (1–4 h)	8+ timepoints (every 30–60 min)
QC Parameters	AV, OHV, moisture	AV (main), GC, moisture, colour	AV, GC, KF, colour, OHV, density, viscosity
Test Frequency	End of batch	Every 1–2 h	Continuous / online + peridic lab checks
Acceptance Criteria	AV 8–10; OHV 140–150; moisture <0.5% GC mono-dominant	Purity >95%, AV 2–5 water <500 ppm	Purity >95–98%; AV 2–4 water <500 ppm, colour < 1.5

EQUIPMENT

SCALES:	LAB	INDUSTRIAL	PILOT
Reaction Vessel	A 2-liter three-neck round-bottom flask	20 L jacketed glass-lined or SS316L reactor	150 L SS316L jacketed batch reactor
Agitation	Magnetic stirrer 300-400 rpm	Overhead agitator (anchor/propeller)-200-400 rpm	3-blade turbine agitator (2kW) 150 rpm
Water Removal	A vacuum distillation head and chilled receiving flask	Overhead Partial Condenser +Phase Separator	Continuous condenser–decanter under vacuum
Temperature Control	A thermocouple with a PID controller	RTD (Pt-100) with PID Controller Temperature Control Valve	Dual RTD (DCS based temp controller) + automated control valves
Vacuum System	A pump with a vacuum gauge	Vacuum pump + moisture trap	Vacuum system with receiver & scrubber
Filtration	pressure filter	Nutsche or cartridge filter SS316L	Sparkler or pressure leaf filter SS316L



HIIRA AND CONTROLS

LAB SCALE:

HIRA:

- Thermal burns from molten reactants and overheated oil baths.
- Corrosive acid catalyst; risk of toxic acrolein formation from overheated glycerol.
- Bumping due to sudden vacuum application.
- Fire risk from organic vapours contacting hot surfaces.

Controls:

- Heat-resistant gloves, face shield, lab coat.
- Gradual vacuum application after 120 °C.
- Operation in a high-flow fume hood.
- Secure, clamped overhead stirring system.

KILO SCALE:

HIRA:

- Cooling failure can cause rapid temperature rise due to larger reaction mass.
- Loss of N_2 blanketing may create a flammable headspace.
- Agitator seal failure can lead to leaks during heating and mixing.
- Overcharging or pump errors can cause significant spills.
- Improper valve position during pressurization can lead to hot liquid spray.
- Blocked condenser may result in pressure buildup.
- Hot, pressurised sampling poses burn risk.

Controls:

- Monitor jacket temperature; provide manual quench and pressure relief.
- Ensure continuous N_2 flow with pressure indication and alarms.
- Use heat–agitation interlocks and monitor agitator load.
- Metered charging with volume calculations and spill containment.
- Double valves and lockout procedures for transfer lines.
- Inspect vents and condensers before each batch.
- Use cooled, pressure-equalised sampling and proper PPE.

PILOT SCALE:

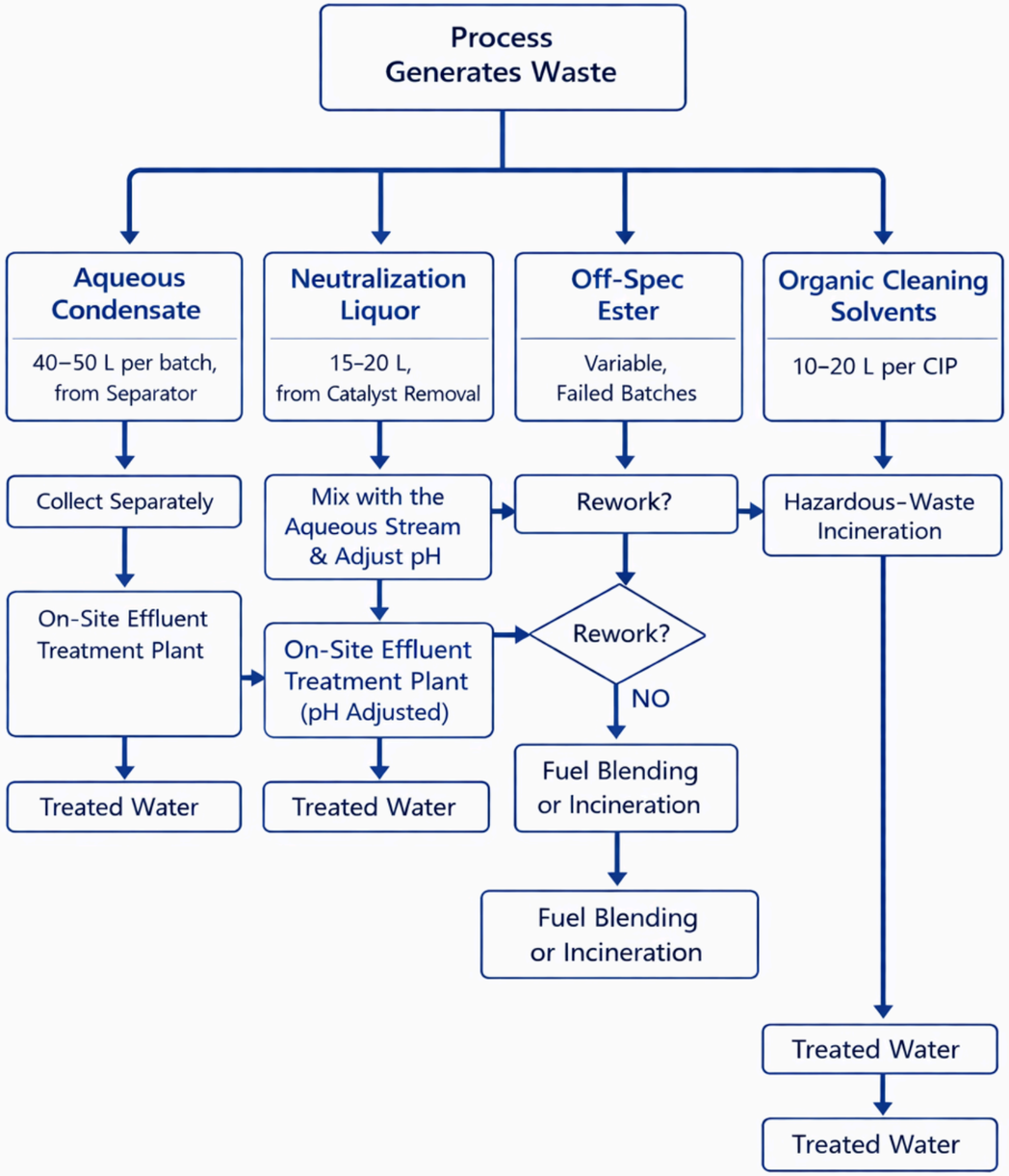
HIRA:

- Loss of cooling or vent blockage can cause temperature and pressure rise.
- Vapour generation may lead to over-pressurisation.
- Flammable vapours may be released during charging or transfer.
- Acid/catalyst splashes pose operator safety risks.
- Seal or gasket failure can cause loss of containment.
- Improper waste handling may cause environmental release.

Controls:

- Temperature alarms with heating cut-off, emergency quench, PSV.
- Nitrogen blanketing, pressure monitoring, PSV to scrubber/vent.
- Closed charging/transfer, ventilation, grounding and bonding.
- Closed acid addition, PPE, eyewash and safety shower.
- Double seals, secondary containment, leak checks, maintenance.
- Dedicated waste tanks and compliant treatment before disposal.

Waste Streams & Environmental Management



Framework for scale - up/down

